

COMPREHENSIVE CANCER CENTER MULTIDISCIPLINARY ONCOLOGY NOTE

Date: 15.03.2024

Patient: Henry Bailey

DOB: 09.03.1967

MRN: AML056

HISTORY OF PRESENT ILLNESS:

Mr. Bailey is a 57-year-old male with adverse-risk acute myeloid leukemia who returns for evaluation of treatment options following disease progression on hypomethylating agent therapy. He was initially diagnosed in October 2022 after presenting with progressive fatigue, dyspnea, and laboratory findings consistent with pancytopenia.

INITIAL WORKUP AND DIAGNOSIS:

Bone marrow biopsy revealed 89% myeloblasts with complex karyotype including del(5q), -7, and +8. Comprehensive molecular profiling identified mutations in EZH2 (p.Y646F, VAF 44%) and STAG2 (frameshift mutation, VAF 39%), along with ASXL1 and SRSF2 mutations. These findings classified his disease as adverse-risk per ELN 2022 guidelines.

TREATMENT HISTORY:

Given his relatively young age but presence of adverse cytogenetics, we initiated therapy with decitabine 15mg/m² IV daily for 10 days of 28-day cycles on October 18th, 2022. He received concurrent oral venetoclax 400mg daily.

Initial response was encouraging with blast reduction to 15% after cycle 2. However, he never achieved complete remission, maintaining 8-12% bone marrow blasts throughout treatment. After 8 cycles, bone marrow assessment in mid June 2023 showed increasing blast percentage (31%) and emergence of additional cytogenetic abnormalities, consistent with progressive disease.

CURRENT STATUS AND PHYSICAL EXAMINATION:

Mr. Bailey reports increased fatigue over the past 2 months but remains functional (ECOG 1). He continues working part-time as an accountant. Physical examination reveals mild pallor, no lymphadenopathy, and no hepatosplenomegaly.

LABORATORY STUDIES (Current):

- Complete blood count: WBC 2.4 (45% blasts), Hemoglobin 8.9 g/dL, Platelets 67,000
- Comprehensive metabolic panel: Within normal limits except creatinine 1.4 mg/dL
- Lactate dehydrogenase: 456 U/L (elevated)
- Bone marrow aspirate and biopsy (March 2024): 58% myeloblasts, complex karyotype with new abnormalities

ASSESSMENT AND PLAN:

Progressive adverse-risk AML following hypomethylating agent failure. Given his

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relatively young age and good performance status, we have discussed several options:

1. **Clinical trial enrollment:** We have identified two potential trials - a FLT3 inhibitor combination study and a novel menin inhibitor protocol. Patient expressing strong interest in experimental therapy.
2. **Salvage chemotherapy:** FLAG-IDA or similar intensive regimen followed by allogeneic stem cell transplant evaluation. Discussed risks given his disease characteristics.

Mr. Bailey will meet with our transplant team next week for preliminary evaluation. We plan to initiate salvage therapy within 2 weeks pending his decision on treatment approach. Supportive care with transfusions and infection prophylaxis will continue.

CURRENT MEDICATIONS:

- Allopurinol 300mg daily
- Levofloxacin 500mg daily
- Fluconazole 200mg daily
- Omeprazole 20mg daily

Next appointment scheduled in 1 week to finalize treatment decision.

Dr. James Rodriguez, MD, PhD
Director, Leukemia Program